Computing Rellich- and Hadamard-Type Integrals for MPS-Derived Eigenfunctions of Planar Domains

Abstract

We present a unified, boundary-integral approach for two families of quantities needed when the Method of Particular Solutions (MPS) is used as a black-box eigensolver in a spectral shape-optimization or inverse-problem pipeline: (i) *Rellich-type* identities, used to L^2-orthonormalize (possibly degenerate) Laplacian eigenfunctions, and (ii) *Hadamard-type* variational formulas, used to differentiate eigenvalues with respect to domain parameters. Both reduce entirely to boundary integrals of Cauchy data, and both integrands are built from the same four bilinear kernels contracted against different scalar weights. We show how the corner-adapted structure already present in the MPS particular-solution basis (finite Fourier–Bessel expansions at each corner, entire fundamental-solution terms elsewhere) allows every boundary integral to be evaluated by a combination of closed-form/series singularity subtraction and plain high-order quadrature, reaching a target precision with a modest, checkable number of node evaluations. We specialize the Hadamard formulas to polygonal and piecewise-B-spline boundary parameterizations, discuss precision management, and lay out a shared computational framework in which Cauchy-data evaluations are computed once and reused across every quantity of interest.

Contents

1 Introduction and scope

An MPS eigensolver, treated as a black box, produces marked eigenvalues λ of the planar Helmholtz/Laplacian eigenproblem on a domain Ω, together with coefficient vectors expressing approximate eigenfunctions in a fixed particular-solution basis. Two further capabilities are needed to turn this into a genuinely useful tool for spectral inverse problems and shape optimization:

(i) An $L^2(\Omega)$-orthonormal basis for each eigenspace (trivial for simple eigenvalues up to normalization, but requiring real work when the eigenvalue is repeated), so that inner products and matrix elements between eigenfunctions are well defined and precisely computable.

(ii) The derivative of each eigenvalue (and, for repeated eigenvalues, the appropriate splitting/mixing matrix) with respect to whatever parameters describe the domain — polygon vertex coordinates, B-spline control points, or otherwise — so that the eigensolver can be used inside a gradient-based optimizer or Newton-type inverse solver.

Both tasks are traditionally approached via area (interior) cubature, which is expensive to make accurate near reentrant corners and is not the natural home for information that MPS already represents on the boundary. This document develops both from a single family of *boundary*-integral identities, exploits the known corner-singularity structure of the MPS basis to reach very high (14–15 digit) precision with few evaluations, and gives a shared software architecture in which the expensive part — evaluating basis functions and their derivatives on the boundary — is done once and reused for every downstream quantity.

1.1 Black-box interface assumed

We assume the MPS solver provides:

- A fixed particular-solution basis $\{\varphi_1, \ldots, \varphi_N\}$, each an exact solution of $\Delta\varphi_j + \lambda\varphi_j = 0$ for a given trial λ, partitioned into:
  - *Corner-singular* terms: for each corner c (interior angle γ_c), a finite set of Fourier–Bessel modes $m = 1, \ldots, M_c$ with M_c fixed in advance by the solver;
  - *Globally smooth* terms: fundamental-solution monopoles/multipoles with sources outside $\overline{\Omega}$, entire throughout Ω.
- The ability to evaluate any φ_j (and its gradient) at an arbitrary user-specified set of points on $\partial\Omega$.
- Coefficient vectors $c_k \in \mathbb{R}^N$, $k = 1, \ldots, K$, for approximate (non-orthogonal) eigenfunctions at marked eigenvalues λ_k (repeated according to estimated multiplicity).

Because point evaluations are the potentially expensive operation, our overriding design goal is to *minimize the number of distinct evaluation points* and to *reuse a single evaluation pass* across every quantity built from it: the Rellich Gram matrix, the Hadamard matrix, and the gradient with respect to every domain parameter.

1.2 Notation summary

$\Omega, \partial\Omega, \mathbf{N}, \mathbf{T}$	domain, boundary, outward normal, unit tangent (ccw)
$\partial_{\mathbf{N}} w, \partial_{\mathbf{T}} w$	$\nabla w \cdot \mathbf{N}$, $\nabla w \cdot \mathbf{T}$
$\mathbf{r}(x) = x - O$	position vector from an arbitrary fixed origin O
λ	Helmholtz eigenvalue, $\Delta u + \lambda u = 0$
γ_c, $p_{c,m}$	interior angle at corner c; admissible radial exponent of mode m there
φ_j	particular-solution basis function
$c_k \in \mathbb{R}^N$	MPS coefficient vector for (unorthonormalized) eigenfunction k
$V(s)$	scalar normal boundary velocity for a shape perturbation
$K^{uv}, K^{NN}, K^{TT}, K^{\mathrm{cr}}$	the four bilinear Cauchy-data kernels (Sec. 7)

2 The Rellich bilinear master identity

Theorem 1 (Master identity)**.** *Let u, v satisfy $\Delta u + \lambda u = 0$, $\Delta v + \lambda v = 0$ in Ω for a common $\lambda \neq 0$. Then*

$$\int_\Omega u\, v\, dA = \frac{1}{2\lambda} \oint_{\partial\Omega} (\mathbf{r}\cdot\mathbf{N}) \Big[\partial_{\mathbf{N}} u\, \partial_{\mathbf{N}} v - \partial_{\mathbf{T}} u\, \partial_{\mathbf{T}} v + \lambda u v\Big] ds + \frac{1}{2\lambda} \oint_{\partial\Omega} (\mathbf{r}\cdot\mathbf{T}) \Big[\partial_{\mathbf{T}} u\, \partial_{\mathbf{N}} v + \partial_{\mathbf{T}} v\, \partial_{\mathbf{N}} u\Big] ds, \quad (1)$$

with no boundary condition assumed.

Sketch. Apply the divergence theorem to $\mathbf{F} = (\mathbf{r}\cdot\nabla u)\nabla v + (\mathbf{r}\cdot\nabla v)\nabla u - \mathbf{r}(\nabla u\cdot\nabla v)$, which satisfies the dimension-independent identity $\nabla\cdot\mathbf{F} = (\mathbf{r}\cdot\nabla u)\Delta v + (\mathbf{r}\cdot\nabla v)\Delta u$. Substituting the PDE and using $\nabla\cdot\mathbf{r} = 2$ in the plane gives $\int_\Omega \nabla\cdot\mathbf{F}\,dA = -\lambda\oint_{\partial\Omega}(\mathbf{r}\cdot\mathbf{N})uv\,ds + 2\lambda\int_\Omega uv\,dA$. Decomposing $\nabla u, \nabla v$ on $\partial\Omega$ into their $\{\mathbf{N},\mathbf{T}\}$ components (valid at every boundary point regardless of any boundary condition) gives $\mathbf{F}\cdot\mathbf{N} = (\mathbf{r}\cdot\mathbf{N})[\partial_\mathbf{N} u\partial_\mathbf{N} v - \partial_\mathbf{T} u\partial_\mathbf{T} v] + (\mathbf{r}\cdot\mathbf{T})[\partial_\mathbf{T} u\partial_\mathbf{N} v + \partial_\mathbf{T} v\partial_\mathbf{N} u]$ on $\partial\Omega$. Equating $\int_\Omega \nabla\cdot\mathbf{F}\,dA = \oint_{\partial\Omega}\mathbf{F}\cdot\mathbf{N}\,ds$ and solving for $\int_\Omega uv\,dA$ gives (1). □

2.1 Specialization by boundary condition

Each boundary condition below is assumed uniform on a given edge; (1) is restricted to that edge and simplified using the stated relation between $u, \partial_\mathbf{N} u, \partial_\mathbf{T} u$.

Dirichlet ($u = 0$). Tangential derivatives vanish identically along the edge, giving

$$\text{contribution} = \frac{1}{2\lambda}\oint_{\Gamma_D}(\mathbf{r}\cdot\mathbf{N})\,\partial_\mathbf{N} u\,\partial_\mathbf{N} v\,ds. \tag{2}$$

Neumann ($\partial_\mathbf{N} u = 0$). All terms containing a normal derivative vanish, giving

$$\text{contribution} = \frac{1}{2\lambda}\oint_{\Gamma_N}(\mathbf{r}\cdot\mathbf{N})\big[\lambda uv - \partial_\mathbf{T} u\,\partial_\mathbf{T} v\big]\,ds \qquad (\lambda\neq 0). \tag{3}$$

Robin, convex-combination form ($(1-a)u + a\,\partial_\mathbf{N} u = 0$, $0 < a < 1$). Writing $\rho := (1-a)/a$, so $\partial_\mathbf{N} u = -\rho u$,

$$\text{contribution} = \frac{1}{2\lambda}\oint_{\Gamma_R}\Big\{(\mathbf{r}\cdot\mathbf{N})\big[(\rho^2+\lambda)uv - \partial_\mathbf{T} u\,\partial_\mathbf{T} v\big] - \rho(\mathbf{r}\cdot\mathbf{T})\,\partial_\mathbf{T}(uv)\Big\}\,ds. \tag{4}$$

The limits $a\to 0, 1$ recover (2), (3) respectively (as a consistency check, not a numerical strategy near those endpoints; edges with a close to 0 or 1 should simply use the Dirichlet/Neumann formula).

Zaremba (mixed). Since $\partial\Omega$ is a closed curve with no boundary of its own, (1) separates *exactly* into the sum of edge contributions above, with no extra junction term — provided each edge integral converges absolutely at its endpoints, which the corner exponents below guarantee.

Remark 1. *Setting $u = v$ recovers the single-function norm identity, e.g. for Dirichlet:* $\displaystyle\int_\Omega u^2\,dA = \frac{1}{2\lambda}\oint_{\partial\Omega}(\mathbf{r}\cdot\mathbf{N})(\partial_\mathbf{N} u)^2\,ds.$

3 Orthonormalization of degenerate clusters

Let λ have estimated multiplicity d, with candidate coefficient vectors $c_1,\ldots,c_d$ (rows of $C\in\mathbb{R}^{d\times N}$).

3.1 Why sandwiching a basis-level matrix between two copies of C is numerically dangerous

A first, mathematically equivalent way to state the Gram matrix is to define the $N \times N$ boundary mass matrix A_{jl} by applying the relevant BC-specialized formula (Sec. 2) to the raw basis pair (φ_j, φ_l), then set $G = CAC^T$. **This construction should be avoided in floating point.** The coefficient matrix C is the solution of a badly conditioned MPS collocation system: its entries are individually large and meaningful only in the specific cancelling combination that makes $u_k = \sum_j c_{kj}\varphi_j$ an accurate eigenfunction. The entries of A, by contrast, are rounded independently at each quadrature node, with no relationship to that cancellation pattern. Forming CAC^T therefore multiplies A's uncorrelated rounding error through C on both sides — the same mechanism by which forming normal equations A^TA squares the effective condition number of a least-squares problem — so that the *absolute* error in G scales like $\|C\|^2\,\delta A$, with $\|C\|^2$ exactly the quantity that is enormous precisely when the basis is ill-conditioned. The true inner products, after the coefficients' cancellation, are typically far smaller than this bound suggests, so G computed this way can be far less orthonormal than the Kress quadrature's own accuracy would predict — exactly the symptom of visibly poor orthonormality despite high-order corner quadrature.

3.2 The safe assembly order: evaluate first, sandwich never

The fix is to perform the coefficient-level cancellation exactly once, at the point it is already known to be benign (the same combination used to evaluate or plot u_k), rather than after being multiplied through an independently-rounded matrix:

1. At each shared boundary quadrature node x_n (Kress-graded near corners, as in Sec. 6), evaluate each candidate eigenfunction's Cauchy data directly from its own coefficients,

$$u_k(x_n) = \sum_{j=1}^{N} c_{kj}\,\varphi_j(x_n), \qquad \partial_{\mathbf{N}}u_k(x_n),\ \partial_{\mathbf{T}}u_k(x_n) \text{ likewise,} \tag{5}$$

   i.e. one stable, direct evaluation per node per candidate — never assembling or storing the $N \times N$ matrix A.

2. Form the kernel values $K(u_k, u_l)(x_n)$ from these already-evaluated numbers.

3. Accumulate with quadrature weights to obtain G_{kl} directly:

$$G_{kl} = \sum_n w_n\, K\big(u_k(x_n), u_l(x_n)\big). \tag{6}$$

Equation (6) is mathematically identical to $G = CAC^T$ but numerically very different: the only place cancellation among the large c_{kj} occurs is step 1, in exactly the combination the MPS solver already trusts for boundary-residual checks and plotting, and every subsequent operation acts on the already-cancelled, well-scaled values $u_k(x_n)$. This replaces the elemental-matrix assembler of Sec. 7 with an evaluate-then-accumulate routine of the same asymptotic cost (still one shared pass over quadrature nodes) but without the squared-conditioning penalty.

Remark 2 (Diagnostic: is the damage upstream or in assembly?)**.** *Before applying the fix, determine which failure mode is present: evaluate $u_k(x_n)$ from* (6)*'s step 1 in ordinary double precision and again in extended (e.g. quadruple) precision at a handful of nodes. If the two agree to full double* *precision, the coefficients themselves are trustworthy for evaluation and the problem is specifically the CAC^T assembly order — Sec. 3.2's fix is sufficient. If they disagree already at this stage, the ill-conditioning has corrupted c_k below the precision needed even for pointwise evaluation, and no downstream reordering can repair it; the coefficients must be recovered more accurately upstream, e.g. by re-solving the MPS collocation system in a preconditioned basis (a QR or SVD-based reconditioning of the boundary-sampled basis evaluation matrix, in the spirit of Betcke and Trefethen's resolving approach) rather than the raw $\{\varphi_j\}$.*

Remark 3 (A cheap internal safeguard)**.** *If step 1's length-N dot product is itself found to lose digits at a given node (plausible for very large N or very strong cancellation), use compensated (Kahan) summation or a quadruple-precision accumulation for that one sum; this is inexpensive since it is a single dot product per node, not a matrix operation.*

Once G is computed safely, diagonalize $G = Q\Lambda Q^T$ and set $D = Q\Lambda^{-1/2}Q^T C$ (Löwdin orthogonalization); the orthonormalized eigenfunctions $v_k = \sum_j D_{kj}\varphi_j$ then satisfy $\int_\Omega v_k v_l \, dA \approx \delta_{kl}$ to the precision at which G was computed via (6). This choice minimizes $\sum_i \|v_i - u_i\|_{L^2}^2$ among all valid orthonormalizations and preserves symmetry-adapted combinations, which matters when the eigenspace is later fed into the Hadamard matrix of Sec. 8.3. Distinct eigenvalues are automatically L^2-orthogonal via Green's second identity (no boundary-integral computation needed), to the precision that each eigenfunction individually satisfies its boundary condition.

Remark 4. *Note that D itself is still a linear recombination of the original, individually large c_{kj}'s — forming $v_k = \sum_j D_{kj}\varphi_j$ for later evaluation (e.g. inside the Hadamard matrix of Sec. 8.3) is safe by the same logic as step 1 above, since it is a single direct evaluation from coefficients, not a sandwich through an independently-rounded matrix. The danger identified in Sec. 3.1 is specific to inserting an independently-assembled matrix* between *two copies of an ill-conditioned coefficient matrix; it does not recur merely because D's entries remain large.*

4 Corner structure of the particular-solution basis

4.1 Admissible exponents

Near corner c (interior angle γ_c), in local polar coordinates (ρ, θ) with the two edges at $\theta = 0, \gamma_c$, admissible Fourier–Bessel modes take the form $\varphi_m(\rho, \theta) = J_{p_m}(\sqrt{\lambda}\rho)\,\Theta_m(\theta)$, with exponent family and angular function fixed by the pair of boundary conditions meeting there:

BC pair at corner	$\Theta_m(\theta)$	p_m
Dirichlet–Dirichlet	$\sin(m\pi\theta/\gamma_c)$	$m\pi/\gamma_c,\ m = 1, 2, \ldots$
Neumann–Neumann, Robin–Robin	$\cos(m\pi\theta/\gamma_c)$	$m\pi/\gamma_c,\ m = 0, 1, 2, \ldots$
Dirichlet–Neumann/Robin (Zaremba)	mixed	$(m + \frac{1}{2})\pi/\gamma_c,\ m = 0, 1, 2, \ldots$

A finite Robin impedance perturbs only the coefficients within a fixed exponent family, not the exponents themselves, so Robin edges share the Neumann exponent family. Terms with p_m a nonnegative integer are entire in ρ (harmless); non-smoothness at $\rho = 0$ occurs precisely when $p_m \notin \mathbb{Z}_{\geq 0}$, which for reentrant corners ($\gamma_c > \pi$) always includes the leading mode.

4.2 The entire-function representation of the radial factor

Using $J_p(z) = (z/2)^p\, {}_0F_1(;p+1;-z^2/4)/\Gamma(p+1)$, define

$$E_p(\rho) := \frac{1}{\Gamma(p+1)}\, {}_0F_1\left(;p+1;-\frac{(k\rho)^2}{4}\right), \qquad k = \sqrt{\lambda}, \tag{7}$$

so that $J_p(k\rho) = (k\rho/2)^p E_p(\rho)$ exactly, with E_p entire (a power series in ρ^2) and evaluable stably at $\rho = 0$ ($E_p(0) = 1/\Gamma(p+1)$), avoiding the cancellation risked by evaluating J_p directly and dividing by $(k\rho/2)^p$. On the boundary ray (θ fixed at the edge value), one obtains

$$\partial_{\mathbf{N}}\varphi_m(\rho) = \pm\Theta'_m(\theta_{\text{edge}}) \left(\tfrac{k}{2}\right)^{p_m} \rho^{p_m-1} E_{p_m}(\rho), \tag{8}$$

$$\partial_{\mathbf{T}}\varphi_m(\rho) = \Theta_m(\theta_{\text{edge}})\, \rho^{p_m-1} \left[k \left(\tfrac{k}{2}\right)^{p_m-1} E_{p_m-1}(\rho) - p_m \left(\tfrac{k}{2}\right)^{p_m} E_{p_m}(\rho)\right], \tag{9}$$

using the contiguous relation $J'_p(z) = J_{p-1}(z) - (p/z)J_p(z)$ for the tangential (radial) derivative. Both expressions are single powers of ρ times entire functions — no numerical differentiation, and no term of the ladder $p_1, p_2, \ldots$ needs to be handled differently in kind, only in exponent.

5 Singularity subtraction for boundary integrals

5.1 The SS/SR/RS/RR decomposition

Near corner c, split every function into the part built from corner-c modes and everything else:

$$u = u_c^{\text{sing}} + u_c^{\text{reg}}, \qquad u_c^{\text{sing}} = \sum_{j\in S_c} c_j\varphi_j, \quad u_c^{\text{reg}} = \sum_{j\notin S_c} c_j\varphi_j. \tag{10}$$

u_c^{reg} is real-analytic near c (every other corner's Fourier–Bessel terms are singular only at their own corner; global terms are entire), but has no closed form — only numerical evaluation, which the MPS evaluator already provides. Any bilinear kernel from Sec. 7 splits into four pieces:

$$K(u,v) = \underbrace{K(u_c^{\text{sing}}, v_c^{\text{sing}})}_{\text{SS}} + \underbrace{K(u_c^{\text{sing}}, v_c^{\text{reg}}) + K(u_c^{\text{reg}}, v_c^{\text{sing}})}_{\text{SR/RS}} + \underbrace{K(u_c^{\text{reg}}, v_c^{\text{reg}})}_{\text{RR}}. \tag{11}$$

SR/RS terms are *not* smooth (one singular factor remains) and must not be lumped with RR.

5.2 Evaluation strategy per piece

RR. Fully smooth; plain high-order Gauss–Legendre.

SS. Each mode pair (m, m') contributes a term $\rho^a \times$ (entire series in ρ^2) for an explicit exponent $a = p_m + p_{m'} - 2$ (kernels with two derivatives) or similar. Two evaluation routes:

- *Series route (recommended default):* multiply the two E-series, giving $\sum_n b_n \rho^{a+2n}$; integrate term by term in closed form, $\int_0^R \rho^{a+2n}\, d\rho = R^{a+2n+1}/(a+2n+1)$ (a log term replaces this only in the measure-zero case $a + 2n + 1 = 0$, cheap to guard against). Truncate at the n dictated by the target precision via the series' own tail bound.

- *Closed-form route:* Lommel's formulas, $\int_0^R zJ_\mu(z)J_\nu(z)\,dz = \frac{R[\mu J_\nu(R)J_{\mu-1}(R) - \nu J_\mu(R)J_{\nu-1}(R)]}{\mu^2-\nu^2}$ ($\mu \neq \nu$) and the analogous $\mu = \nu$ formula, exact with zero truncation error, at the cost of more bookkeeping to reduce every derivative combination to this exact form.

SR/RS. For each mode m, u_c^{sing}'s contribution is $\rho^{p_m-1} \times$ (entire); multiplied by the smooth, numerically-evaluated v_c^{reg} (or its derivative), this is integrated with a Gauss–Jacobi rule tuned to exponent $p_m - 1$ (or p_m for value-type kernels), with the entire/smooth factors evaluated at the rule's nodes.

5.3 Corner exponents for Hadamard-relevant kernels

The same exponent bookkeeping applies to kernels K^{uv} (leading exponent $2p_1$, using the leading *non-smooth* mode, i.e. skipping an analytic $m = 0$ term where present), K^{NN}, K^{TT} ($2(p_1 - 1)$), and K^{cr} ($2p_1 - 1$), with p_1 read off Sec. 4's table for whichever BC pair actually meets at that point — including Zaremba junctions with no geometric corner ($\gamma = \pi$), which nonetheless force half-integer exponents.

Remark 5. *The SS/SR/RS/RR decomposition above gives an explicit, checkable error certificate on the singular part of every integral, but at a real practical cost: it requires a separate Gauss–Jacobi rule (hence a separate set of evaluation points for u_c^{reg}) per retained mode $m = 1, \ldots, M_c$ at each corner, and every one of those points requires a fresh Bessel/hypergeometric evaluation of every other basis function. Where this cost is prohibitive — e.g. many corners, many retained modes, many eigenfunctions per shape-optimization step — Sec. 6 presents two alternatives that use a single shared quadrature rule per corner panel, applied directly to the raw (undecomposed) eigenfunction, at the cost of losing this exact error certificate on the singular part.*

6 Alternatives: single-rule quadrature for corner panels

The decomposition of Sec. 5 gives the strongest error guarantees but requires a new evaluation pass (and hence new Bessel/hypergeometric calls against every other basis function) for each retained mode at each corner. When this evaluation cost dominates — as is common once several corners, several modes, and many eigenfunctions per optimization step are all in play — it is often preferable to trade the exact certificate on the singular part for a *single* quadrature rule per corner panel, applied directly to the raw Cauchy data of the full eigenfunction (no SS/SR/RS/RR split at all). We describe two such approaches.

6.1 Kress-style graded-mesh quadrature

The classical approach to algebraic endpoint singularities in boundary integral equation practice (Kress, *A Nyström method for boundary integral equations in domains with corners*, SIAM J. Numer. Anal. 1990; see also Colton & Kress, *Inverse Acoustic and Electromagnetic Scattering Theory*) is to reparametrize the corner-adjacent panel by a fixed, smooth *grading map* $w : [0,1] \to [0,1]$ chosen so that w and a prescribed number of its derivatives vanish at $t = 0$ (and, by the symmetric construction $w(t) = 1 - w(1-t)$, so does $1 - w$ at $t = 1$). A standard sigmoidal form used for this purpose is

$$w(t) = \frac{v(t)^q}{v(t)^q + v(1-t)^q}, \qquad v(t) = \left(\frac{1}{q} - \frac{1}{2}\right)(1-2t)^3 + \frac{1}{q}(2t-1) + \frac{1}{2}, \tag{12}$$

for an integer grading order $q \geq 2$, chosen so that $w^{(k)}(0) = 0$ for $k = 1, \ldots, q-1$.

Remark 6 (Confidence note)**.** *Equation* (12) *reflects the standard qualitative construction (a polynomial sigmoid flattening derivatives to a prescribed order at both endpoints) attributed to Kress; the* *exact coefficients above should be checked against the primary source before being hard-coded, and verified directly by symbolic or numerical differentiation ($w^{(k)}(0) \stackrel{?}{=} 0$ for $k < q$) before production use.*

Composing the panel's arclength parametrization with w clusters quadrature nodes near the corner geometrically (in effect) while remaining a single smooth substitution rather than an explicit set of dyadic sub-panels; a fixed-order Gauss–Legendre rule applied in the new variable t then integrates the composed integrand — singular part times grading Jacobian — with algebraic convergence order controlled by q relative to the true exponent $p_1 - 1$ appearing in the kernel. Convergence is not exponential (unlike the closed-form SS route or the RR panels elsewhere in this document), but for practical q (commonly 6–10) the rate is fast enough to reach a 14–15 digit target with a modest, fixed node count once q is chosen comfortably larger than the worst exponent present.

Why this helps with evaluation cost. No mode-by-mode split is needed: the same graded node set is used to evaluate the *raw* eigenfunction's full Cauchy data once, and that single evaluation serves every kernel ($K^{uv}, K^{NN}, K^{TT}, K^{\mathrm{cr}}$) and every weight ($\mathbf{r}\cdot\mathbf{N}, \mathbf{r}\cdot\mathbf{T}, V$) at that corner, directly addressing the Bessel-call bottleneck. A simpler, less refined alternative in the same spirit — geometric/dyadic panel refinement toward the corner with ordinary Gauss–Legendre on each sub-panel, rather than one smooth graded map — achieves a similar qualitative effect with less setup but introduces explicit panel joints (a mild reduction in achievable order at each joint) and is worth considering as a lower-effort first attempt before implementing (12).

Certification. Since the exact SS error bound is given up, verify empirically: increase q (or the panel count, for dyadic refinement) and confirm the target kernel values stabilize to the required digit count, exactly as for the RR panels in Sec. 10.

6.2 Ma–Rokhlin–Wandzura generalized Gaussian quadrature

A more aggressive option, in the spirit of the Vioreanu–Rokhlin-type rules already used for smooth panels elsewhere in this pipeline, is to construct a *generalized Gaussian quadrature* rule tailored exactly to the known function set present at a given corner (Ma, Rokhlin, Wandzura, *Generalized Gaussian quadrature rules for systems of arbitrary functions*, SIAM J. Numer. Anal. 34 (1996)).

Construction. Given the $2n$ functions $\{\rho^{p_1-1}, \dots, \rho^{p_{M_c}-1}\} \cup \{1, \rho, \dots\}$ (the corner's known singular exponents together with enough low-degree polynomials to represent the smooth remainder to the desired order), seek n nodes t_i and weights w_i satisfying

$$\sum_{i=1}^{n} w_i\, f_k(t_i) = \int_0^R f_k(\rho)\, d\rho \qquad \text{for } k = 1, \dots, 2n, \tag{13}$$

a nonlinear system in $2n$ unknowns, solved by Newton's method from a suitable initial guess (e.g. continuation from a nearby corner angle, or from an ordinary Gauss–Jacobi rule matched to the leading exponent). The right-hand sides — the moments of each f_k — are exactly the closed-form/series integrals already derived in Sec. 5 for the SS pieces, now reused as *input data* to build the rule rather than as a final answer.

Advantages. The resulting n-node rule is exact (to the precision of the Newton solve) for every function in the set simultaneously, typically needing substantially fewer nodes than a graded polynomial rule for the same target precision, since it is tailored to the exact singular functions rather than approximating them. At runtime it behaves exactly like an ordinary quadrature rule: evaluate the raw eigenfunction's Cauchy data at the n nodes, dot with the n weights — one evaluation pass per corner panel, shared across every kernel and weight, same as the graded-mesh approach but typically with fewer total nodes.

Caveats specific to a shape-optimization setting. The rule depends on the corner angle γ_c (through the exponents p_m) as well as the mode list M_c, both fixed for a static domain but not for a domain being continuously deformed during optimization or inversion. Two practical responses: (i) rebuild the rule at each optimization step, warm-started from the previous step's node/weight values, which is typically a cheap Newton solve given a good starting point and a small angle change; or (ii) build the rule once for a nominal angle with a modest robustness margin and empirically verify (via the same doubling/stability check as elsewhere) that accuracy remains acceptable as the angle drifts within the optimizer's trajectory, falling back to a rebuild only if it does not. The nonlinear solve underlying the construction is a well-trodden but nontrivial numerical exercise in its own right; see also the related corner-quadrature constructions of Bremer and Rokhlin for Helmholtz layer potentials for implementation guidance.

6.3 Comparison

	SS/SR/RS/RR (Sec. 5)	Kress graded mesh	MRW generalized quadrature
Error certificate	exact (series/closed-form)	empirical only	empirical only
Evaluation passes per corner	one per mode m	one, shared	one, shared
Setup cost	none beyond series/Lommel data	low (fixed map)	higher (nonlinear solve)
Node count for target precision	typically fewest	moderate	typically fewer than Kress
Robust to angle drift in optimization	yes (exponents recomputed directly)	yes	needs rebuild or margin

The right choice depends on which resource is scarcest: SS/SR/RS/RR when evaluation calls are cheap and an exact certificate is valuable; Kress grading as a simple, robust default when evaluation cost dominates and angles vary during optimization; MRW quadrature when the same corner geometry is reused across very many eigenfunctions (amortizing its setup cost) and the angle is either fixed or changes slowly enough to tolerate periodic rebuilding.

7 A unified computational framework

Both Rellich and Hadamard integrands are linear combinations of four bilinear Cauchy-data kernels,

$$K^{uv} = uv, \quad K^{NN} = \partial_{\mathbf{N}} u \, \partial_{\mathbf{N}} v, \quad K^{TT} = \partial_{\mathbf{T}} u \, \partial_{\mathbf{T}} v, \quad K^{\mathrm{cr}} = \partial_{\mathbf{T}} u \, \partial_{\mathbf{N}} v + \partial_{\mathbf{T}} v \, \partial_{\mathbf{N}} u, \tag{14}$$

contracted against a scalar weight ($\mathbf{r} \cdot \mathbf{N}$ or $\mathbf{r} \cdot \mathbf{T}$ for Rellich; the normal boundary velocity $V(s)$ for Hadamard). This suggests a three-layer architecture, structured specifically to avoid the ill-conditioning hazard identified in Sec. 3.1.

1. **Raw basis Cauchy data (once per basis function, per shared node set).** For each φ_j, evaluate $(\varphi_j, \partial_\mathbf{N}\varphi_j, \partial_\mathbf{T}\varphi_j)$ at a single, shared set of boundary quadrature nodes per edge/panel — chosen dense enough (with the Kress or MRW treatment of Sec. 6 applied at corner-adjacent panels) to serve *every* downstream quantity, since none of the weight functions used below change what data is needed from the basis functions themselves. This layer involves no ill-conditioned quantities and can be cached freely.

2. **Eigenfunction Cauchy data (once per eigenfunction, per shared node set).** At the same nodes, evaluate each eigenfunction of interest directly from its own coefficient row,

$$u_k(x_n) = \sum_j c_{kj}\varphi_j(x_n), \qquad \partial_\mathbf{N} u_k(x_n),\ \partial_\mathbf{T} u_k(x_n) \text{ likewise}, \tag{15}$$

using C for candidate (pre-orthonormalization) eigenfunctions or D for the Löwdin-orthonormalized cluster (Sec. 3). This is a single dot product per node per eigenfunction — exactly step 1 of Sec. 3.2 — and is the *only* place the ill-conditioned coefficients are used; no basis-level $N \times N$ matrix is ever formed from them.

3. **Generic kernel-and-weight accumulator.** A routine `assemble(kernel, weight, eigfns)` that forms K^{kernel} pointwise from the already-evaluated eigenfunction Cauchy data of layer 2 and accumulates $\sum_n w(x_n)\, K^{\text{kernel}}(x_n)$ with the quadrature weights, directly producing the small (cluster-size, not N-size) Gram or Hadamard matrix entry. Every formula in this document — Rellich (Sec. 2), Hadamard (Sec. 8) — is a fixed linear combination of calls to this one routine with different weights and different eigenfunction pairs.

Remark 7. *This differs from a naive reading of "assemble a basis-level matrix, then multiply by coefficients on both sides" precisely where that reading is dangerous: layer 3 never receives a raw $N \times N$ matrix to sandwich, because layer 2 has already collapsed each eigenfunction's Cauchy data to a single, well-scaled numerical value per node before any kernel or weight is applied. Layer 1's raw basis data is used only to* construct *layer 2 (a benign, single dot product), never combined with a second copy of C or D afterward.*

Layer 1 remains the most expensive step (basis-function evaluation, including any Bessel/hypergeometric calls), and it does not depend on which weight, which eigenfunction, or which downstream formula will use it — so it is still computed once and reused for the Rellich Gram matrix, the Hadamard matrix, and the gradient with respect to every domain parameter (Sec. 9). Layer 2 is comparatively cheap (one dot product per eigenfunction per node) and is recomputed whenever the coefficient matrix in use changes (e.g. $C \to D$ after orthonormalization), rather than cached across that transition.

8 Hadamard-type variational formulas

8.1 Setup: normal boundary velocity

A one-parameter family of domains Ω_ε is generated by displacing each boundary point by $\varepsilon V(s)\mathbf{N}(s)$, where V is the scalar normal velocity (extended into a full vector field, e.g. $\varepsilon V\mathbf{N}$, only its normal component matters for first-order eigenvalue perturbation — this is the classical Hadamard structure theorem, which we do not re-derive here). Since MPS basis functions are given by globally defined analytic formulas (Fourier–Bessel and fundamental-solution expressions), rather than functions only defined on Ω itself, they extend automatically a small distance across $\partial\Omega$; this sidesteps the usual technical difficulty of constructing a boundary extension in shape calculus.

8.2 Simple Dirichlet eigenvalues: derivation

Theorem 2. *Let λ be simple with L^2-normalized eigenfunction u ($\int_\Omega u^2\, dA = 1$), Dirichlet condition on $\partial\Omega$. Then*

$$\lambda'(0) = -\oint_{\partial\Omega} V\, (\partial_{\mathbf{N}} u)^2\, ds. \tag{16}$$

Derivation. Let $u_\varepsilon, \lambda_\varepsilon$ be the eigenpair on Ω_ε ($\int_{\Omega_\varepsilon} u_\varepsilon^2 = 1$), and assume $V \geq 0$ so $\Omega \subset \Omega_\varepsilon$ (the general sign follows by linearity in ε). Green's second identity on Ω (where both u, u_ε are defined):

$$\int_\Omega (u\Delta u_\varepsilon - u_\varepsilon \Delta u)\, dA = \oint_{\partial\Omega} (u\, \partial_{\mathbf{N}} u_\varepsilon - u_\varepsilon\, \partial_{\mathbf{N}} u)\, ds. \tag{17}$$

The left side equals $(\lambda - \lambda_\varepsilon)\int_\Omega u\, u_\varepsilon\, dA$. On the right, $u \equiv 0$ on $\partial\Omega$, leaving $-\oint_{\partial\Omega} u_\varepsilon\, \partial_{\mathbf{N}} u\, ds$. Since $u_\varepsilon = 0$ on the displaced boundary $\partial\Omega_\varepsilon$, Taylor-expanding along the normal gives, at each $x \in \partial\Omega$,

$$0 = u_\varepsilon(x + \varepsilon V(x)\mathbf{N}(x)) = u_\varepsilon(x) + \varepsilon V(x)\, \partial_{\mathbf{N}} u_\varepsilon(x) + O(\varepsilon^2), \tag{18}$$

so $u_\varepsilon(x) = -\varepsilon V(x)\partial_{\mathbf{N}} u_\varepsilon(x) + O(\varepsilon^2) = -\varepsilon V(x)\partial_{\mathbf{N}} u(x) + O(\varepsilon^2)$ (using $\partial_{\mathbf{N}} u_\varepsilon \to \partial_{\mathbf{N}} u$ as $\varepsilon \to 0$, standard analytic perturbation theory for a simple eigenvalue). Substituting,

$$(\lambda - \lambda_\varepsilon)\int_\Omega u\, u_\varepsilon\, dA = \varepsilon \oint_{\partial\Omega} V\, (\partial_{\mathbf{N}} u)^2\, ds + O(\varepsilon^2). \tag{19}$$

As $\varepsilon \to 0$, $\int_\Omega u u_\varepsilon\, dA \to 1$ and $\lambda - \lambda_\varepsilon = -\varepsilon\lambda'(0) + O(\varepsilon^2)$; equating the $O(\varepsilon)$ terms gives (16). □

Remark 8. *The corner points contribute a set of measure zero to the boundary integral and require no special treatment in this derivation, regardless of whether the perturbation also moves the corner itself or changes its angle — this only affects u_ε for $\varepsilon \neq 0$, which the first-order formula does not need.*

8.3 Degenerate clusters: matrix generalization

For a multiplicity-d eigenvalue with L^2-orthonormal basis $u_1, \dots, u_d$ (Sec. 3), the direct bilinear generalization of (16) is the symmetric matrix

$$M_{ij} := -\oint_{\partial\Omega} V\, \partial_{\mathbf{N}} u_i\, \partial_{\mathbf{N}} u_j\, ds. \tag{20}$$

As in standard (Rayleigh–Schrödinger) degenerate perturbation theory, the eigenvalues of M give the first-order splitting of λ under the perturbation, and its eigenvectors give the linear combinations of $u_1, \dots, u_d$ that vary smoothly with ε (i.e. the correct zeroth-order eigenfunctions to use for any subsequent, e.g. second-order, perturbation analysis). This is consistent with (16) on the diagonal and is symmetric by construction, matching the pattern by which the Rellich norm identity generalized to the bilinear Gram matrix in Sec. 3.

8.4 Neumann and Robin cases

The analogous Neumann formula (simple eigenvalue, $\int u^2 = 1$, $\partial_{\mathbf{N}} u = 0$) is classically

$$\lambda'(0) = \oint_{\partial\Omega} V \left[(\partial_{\mathbf{T}} u)^2 - \lambda u^2 \right] ds, \tag{21}$$

attributed to Hadamard's original 1908 work and given a rigorous treatment by Garabedian and Schiffer (1952); see also Henrot and Pierre, *Shape Variation and Optimization* (2018), for a modern derivation. Unlike the Dirichlet case, deriving (21) requires tracking how the boundary normal itself rotates under the perturbation (a curvature/tangential-derivative-of-V effect) and eliminating a second normal derivative of u via the PDE and an along-boundary integration by parts — involved enough that we do not reproduce the full derivation here, and instead cite the result.

Remark 9 (Confidence and recommended validation)**.** *Equation* (21) *is a cited classical result, not independently re-derived in this document; a plausible Robin analogue (by the same $\rho = (1-a)/a$ substitution pattern used in* (4)*) would be*

$$\lambda'(0) \stackrel{?}{=} \oint_{\partial\Omega} V \left[(\partial_{\mathbf{T}} u)^2 - (\rho^2 + \lambda) u^2 \right] ds, \tag{22}$$

but this has not *been independently derived or verified here and should be treated as conjectural pending confirmation. Before relying on* (21) *or* (22) *inside an optimizer or inverse solver, we recommend a direct numerical check: recompute the eigenvalue by re-running the MPS solver on an explicitly perturbed domain Ω_ε for a small ε, and compare $(\lambda_\varepsilon - \lambda)/\varepsilon$ against the boundary-integral prediction, for a handful of test perturbations. A sign or coefficient error here is exactly the kind of mistake that would be invisible in isolated unit tests but corrosive inside an optimization loop.*

8.5 Zaremba boundaries

As with the Rellich identity, the total Hadamard integral is the sum of (16)-type contributions over Γ_D and (21)-type contributions over Γ_N (or their Robin analogues over Γ_R), with no separate junction term, subject to the same caveat that the boundary velocity V can itself be non-smooth exactly at a BC-type junction even when the geometric boundary is straight there (Sec. 9).

9 Reduction to concrete domain parameterizations

9.1 Polygonal domains: vertex velocity fields

Perturbing vertex P_j by $\varepsilon\hat{\mathbf{e}}$ ($\hat{\mathbf{e}} = \hat{x}$ or $\hat{y}$), holding neighbors $P_{j\pm1}$ fixed, induces an exactly affine velocity on each of the two adjacent edges: parameterizing edge (P_{j-1}, P_j) by $t \in [0,1]$ ($t = 0$ at P_{j-1}),

$$V_j^{\hat{\mathbf{e}}}(t) = t\,(\hat{\mathbf{e}} \cdot \mathbf{N}_{j-1,j}), \tag{23}$$

and symmetrically $V_j^{\hat{\mathbf{e}}}(t) = (1-t)(\hat{\mathbf{e}} \cdot \mathbf{N}_{j,j+1})$ on edge (P_j, P_{j+1}), zero elsewhere. This is polynomial of degree 1 — exactly as smooth as $\mathbf{r} \cdot \mathbf{N}$ in the Rellich identity — so it requires no new quadrature treatment beyond the corner machinery already applied at P_j itself. Each edge contributes only two independent shapes (ramp-up, ramp-down), reusable across all four vertex-coordinate gradients ($P_j^x, P_j^y, P_{j+1}^x, P_{j+1}^y$) by a scalar multiplier, so per-edge assembly cost is shared across every vertex gradient touching that edge.

9.2 Piecewise B-spline boundaries: control-point velocity fields

Perturbing control point Q_k of one spline piece by $\varepsilon\hat{\mathbf{e}}$ induces

$$V_k^{\hat{\mathbf{e}}}(t) = N_k(t)\,(\hat{\mathbf{e}}\cdot\mathbf{N}(t)), \tag{24}$$

where N_k is the spline's own (compactly supported, polynomial) blending function. Consequences:

- *Locality:* N_k is nonzero only on a few knot spans, so only those spans require reassembly; for plain (non-shared-control-point) B-spline pieces, a perturbation never touches a neighboring piece unless an explicit continuity constraint ties their control points together, in which case the gradient sums contributions from both sides of the constraint.
- *Interior knots:* simple interior knots are geometric C^{p-1} points — no fractional-exponent singularity, hence no corner machinery needed — but still warrant a panel boundary there for full spectral quadrature convergence, since V, $\mathbf{N}(t)$, and ds/dt are only piecewise-polynomial in t, not entire.
- *Explicit junctions* (reduced-multiplicity knots representing a true corner): full corner machinery (Sec. 4–5) applies, exactly as at a polygon vertex.
- *Which control point is perturbed does not matter for validity:* Q_0 (moves the junction point), Q_1 (moves the tangent/angle there), and $Q_2, Q_3, \ldots$ (affect only curvature and beyond) all induce a V that is smooth on each span and evaluated (zero or not, to the corresponding order) at the junction; the first-order formula only ever needs the *unperturbed* eigenfunction's Cauchy data, so none of these require special treatment beyond the ordinary corner quadrature already used at that junction.

10 Precision management

Two structurally different sources of error are in play, and only one of them presents a genuine cost/precision tradeoff.

10.1 Where full precision is essentially free

RR panels (Sec. 5) and any edge where the weight function is polynomial ($\mathbf{r}\cdot\mathbf{N}, \mathbf{r}\cdot\mathbf{T}$ on a polygon edge; V on a polygon edge) are smooth, and high-order Gauss–Legendre reaches double/quad precision with a modest, fixed number of nodes — doubling the node count and confirming digit stability is sufficient certification; no design decision is required here.

10.2 The genuine tradeoff: corner panel radius vs. mode truncation vs. SR convergence rate

Three quantities interact at a corner-adjacent panel of radius R_c:

- *Mode truncation* M_c for the SS piece: governed by the decay rate of the Fourier–Bessel series (a checkable tail bound, generally already estimated from MPS coefficient decay), fewer modes needed for smaller R_c.
- *Gauss–Jacobi convergence rate for SR/RS pieces:* governed by the analyticity radius (in the Bernstein-ellipse sense) of the numerically-evaluated smooth factor u_c^{reg}, which shrinks as R_c approaches the distance to the nearest *other* singularity (another corner, or a BC-type junction). Smaller R_c relative to that distance improves this convergence rate.

- *Total panel count away from the corner:* shrinking R_c pushes more of the edge into the (cheap, smooth) RR/plain-Legendre regime but requires more panels there if graded refinement is used to approach R_c smoothly.

A reasonable default is to set R_c to some modest fraction (e.g. one third to one half) of the distance from corner c to the nearest other singular point on the same edge, choose M_c from the series tail bound at that R_c for the target precision, and verify by the doubling test below. This is the one place in the pipeline where a poor choice can leave you a few digits short of target without an obvious symptom other than a stalled convergence curve, so it is worth the one-time calibration.

Remark 10. *This three-way tradeoff is specific to the exact SS/SR/RS/RR route. If evaluation cost (Bessel/hypergeometric calls) rather than quadrature design is the binding constraint, Sec. 6's single-rule alternatives trade this calibration problem for a simpler one (grading order q, or generalized-quadrature node count n), at the cost of an empirical rather than exact error certificate.*

10.3 Boundary residual: the true precision floor

Every identity in this document requires u, v (or their required Cauchy data) to satisfy the boundary condition *exactly*; MPS coefficients satisfy it only up to the residual left by the boundary collocation solve. This — not quadrature — is generally the binding constraint on achievable precision, and the standard remedy is to recompute eigenfunction coefficients with a higher-quality set of collocation points once an eigenvalue has been located, targeting the same precision as the eigensolver's own stated tolerance (there is no benefit to pushing quadrature past the precision already lost to boundary residual, nor is there benefit to targeting fewer digits than the solver already achieves elsewhere).

10.4 Certification

After assembling any elemental matrix $A^{\text{kernel}}(w)$, recompute it with doubled quadrature order on corner panels and confirm digit stability; separately, verify each eigenfunction's boundary sup-norm is at the target residual level before trusting downstream Gram or Hadamard matrices to that precision.

11 Summary of the computational procedure

1. Fix a shared set of boundary quadrature nodes per edge, with graded/corner panels (radius R_c, or a Kress/MRW rule per Sec. 6, calibrated per Sec. 10) at every geometric corner and BC-type junction.

2. Evaluate raw basis Cauchy data $(\varphi_j, \partial_{\mathbf{N}}\varphi_j, \partial_{\mathbf{T}}\varphi_j)$ at these nodes once (Sec. 7, layer 1).

3. For each candidate eigenfunction, evaluate its own Cauchy data directly from its coefficient row (layer 2); **never** assemble a basis-level $N \times N$ matrix and sandwich it between two copies of an ill-conditioned coefficient matrix (Sec. 3.1).

4. Build the Rellich Gram matrix G per degenerate cluster via the evaluate-then-accumulate formula (6) and Löwdin-orthonormalize (Sec. 3), re-evaluating eigenfunction Cauchy data from the new coefficients D before reuse.

5. Build the Hadamard matrix (20) (or its Neumann/Robin analogue, numerically validated) the same way, using the orthonormalized cluster's directly-evaluated Cauchy data and the appropriate velocity field (Sec. 9) for each domain parameter.

6. Certify via node-doubling, boundary-residual checks, and the extended-precision evaluation diagnostic of Sec. 3.2 (Sec. 10) before use in an optimizer or inverse solver.

12 Outlook: spectral inverse problems and shape optimization

With both eigenvalue gradients and orthonormalized eigenfunctions available at machine-comparable precision from purely boundary data, the MPS eigensolver becomes a differentiable map from domain parameters (vertex or control-point coordinates) to spectral data (eigenvalues, and, via the degenerate Hadamard matrix, the correct splitting directions of repeated eigenvalues). This is precisely the interface required by gradient-based shape optimizers (targeting, e.g., extremal eigenvalues or spectral gaps subject to geometric constraints) and by Newton-type spectral inverse solvers (recovering domain parameters from a target spectrum), both of which otherwise require finite-difference eigenvalue derivatives at a substantial accuracy and cost penalty relative to the boundary-integral approach developed here.

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